Practical 5: Sediment transport and formation of placer deposits (diamonds and gold)

Today's activities are designed to better understand the sediment transport principles underlining formation of sedimentary placer deposits, like diamond, gold, but also other heavy minerals like ilmenite, which is a major source of the metal titanium. These principles were first introduced in the week 3 lecture.

Activity 1 – Stoke's Law

Use the two different types of glass cylinders to investigate how Stoke's Law governs settling of sediments. Shake the cylinder with coarse-grained sediment carefully to get sediment into suspension, and then let the cylinder stand on the table for a few seconds. What is the vertical grain size profile once all sediment has come to rest? Can you explain this grain size profile using Stoke's Law?

Secondly, stir the cylinder with the magnetite-quartz mixture? What is the vertical variation in grain size or mineralogy? How can you explain it?

Activity 2 – Stoke's Law in "plunge pools"

Plunge pools are circular depressions in the river bed (typically cut into harder rock forming the river bed) where heavy grains carried by the river can be trapped due to gravity. This activity simulates the processes in such a pool. In this experiments we use magnetite as the placer mineral.The sediment you will use has 10% by volume magnetite and 90% by volume sand.

1. Fill the cup about half way with mixed sand and magnetite.
2. Place the cup in the plastic box and under the tap of the aspirator bottle (or use a pitcher).
3. Fill the glass from the tap or pitcher until lots of grains are "dancing" in the water and some are coming over the side.
4. Watch and describe the movement of the grains.
5. Continue for a few minutes until about half the sediment has come over the top.
6. Remove the cup from the box and carefully decant the water from both by pouring it into the other container. Try not to lose sediment.
7. Use the chart to estimate the percentage of magnetite that stayed in the cup and of magnetite that escaped and was caught in the box.

Discuss the processes behind your observations.

Activity 3 – Flume tank

In this activity we will work in a flume tank to observe the basic processes of sediment transport. The tank is filled with a loose sediment (quartz sand in our case). A pump creates a flow of water over that sediment bed. Pump speed and flume inclination control the velocity of this current. The Hjulstrom diagram predicts that as current velocity increases at constant grain size, the sand grains will be entrained above the threshold velocity. This threshold velocity increases with grain size. Once entrained, grains will move by rolling, saltation and in suspension. Flow over the sediment bed also creates sedimentary bedforms (e.g. ripples, see EART27201 for those of you who are taking that unit).

Increase pump speed and flume inclination (using the metal wheel) to increase flow velocity. Do manipulate the dials for the flume pump carefully. If you want, get one in your group to make a video of the experiment. As flow velocity increases, make observations:

- what sediment transport processes occur?
- what bedforms develop?

Think about the processes responsible for bedform formation.

Activity 4 – Flume tank

This activity is based on activity 3, but it simulates the segregation of quartz and heavy minerals to form fluvial placers. The tank is filled with a mixture of 10% magnetite (density 7.9 g/cm^3) and 90% quartz (density 2.65 g/cm^3). By flowing water over this bed and increasing flow velocity as in activity 2, we will see the formation of bedforms and the segregation of magnetite from quartz to form a placer deposit.

Do manipulate the dials for the flume pump carefully. If you want, get one in your group to make a video of the experiment. As flow velocity increases, make observations:

- once quartz and magnetite segregate, where does the magnetite occur?

Think about the processes responsible for heavy mineral separation and placer deposition.